

MARKET REGIME DETECTION AND SWITCHING RISK

A Project Report Submitted
in Partial Fulfilment of the Requirements

for the Degree of
MASTER OF SCIENCE

in

Mathematics

by

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to the

DEPARTMENT OF MATHEMATICS

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April 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled “**Market Regime Detection And Switching Risk**” submitted by **Atish Motiram Sawant** (Roll Number: 242123104) to the Department of Mathematics, Indian Institute of Technology Guwahati, towards partial fulfilment of the requirements for the degree of Master of Science in Mathematics, has been carried out by him under my supervision.

It is also certified that, along with literature survey, computational implementations have been carried out by the student under the project.

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ABSTRACT

This project applies unsupervised machine learning techniques, namely K-Means clustering and Hidden Markov Models, to detect hidden market regimes in financial time series. Using S&P 500 data from 2001 to 2025, the study identifies distinct market states based on returns, volatility, and drawdown measures. The detected regimes are then used to construct a simple investment strategy that adjusts exposure based on market conditions. The findings demonstrate that regime-aware investing can effectively manage risk and improve decision-making compared to passive approaches. This work highlights the value of AI-based methods for adaptive financial analysis and portfolio management.

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Chapter 1

Introduction

Financial markets play a crucial role in modern economies through the trading of assets such as stocks, bonds, and other securities. Market indices, such as the S&P 500, SENSEX, DAX, FTSE, NIKKEI, etc. are often used as representative metrics for the overall performance of large market segments.

Unlike deterministic systems studied in classical mathematics, financial markets are characterized by uncertainty and randomness. Asset prices evolve as new information, such as economic announcements, political events, and shifts in investor expectations, becomes available. Because such information arrives unpredictably, price movements exhibit stochastic behaviour, meaning randomness plays a fundamental role in financial dynamics.

Empirical observations reveal several notable features in financial time series. Returns typically fluctuate around a stable average, while volatility varies significantly across different time periods. Markets frequently experience phases of relative stability followed by turbulence and large price movements, indicating that financial risk is not constant but evolves over time.

Understanding how market behaviour changes is therefore essential for investors and researchers. The presence of distinct phases with unique statistical characteristics

motivates the study of market regimes. This concept forms the foundation for the modelling approach developed in this project for both investors as well as firms, thereby enabling the latter to raise capital and these lead to price discovery and liquidity.

1.1 Market Regimes

Financial markets do not exhibit uniform behaviour over time but instead pass through distinct phases known as **market regimes**, which are periods during which variables such as returns, volatility, and correlations remain relatively stable before transitioning to another state. The common examples of market regimes are:

- **Growth regimes:** Positive returns, low volatility during economic expansion
- **Crisis regimes:** Declining prices, high volatility during turmoil
- **Transitional regimes:** Brief adjustment periods between stable states

A fundamental challenge is that market regimes are **not directly observable**. Consequently, investors cannot precisely determine when a regime shift occurs simply by watching prices. Instead, regimes must be inferred from observable data, such as returns and volatility patterns.

Understanding regimes is valuable because strategies that perform well during stable conditions may fail during stressed market conditions. Identifying these hidden structures in financial time series is therefore essential for risk management and adaptive investment decisions.

1.2 Investor Risk and Drawdowns

Financial markets inherently involve uncertainty, exposing investors to varying levels of risk over time. One of the most important concerns in portfolio management is the

occurrence of **drawdowns**, which capture sustained losses experienced during adverse market movements.

Definition 1.2.1 (Drawdown). A **drawdown** measures the decline in portfolio value from a historical peak to a subsequent trough before recovery occurs.

Example 1.2.2. Consider an investor whose portfolio grows from 10,000 to 12,000 and later declines to 9,000 before recovering. The drawdown equals 3,000, corresponding to a loss of 25% from the peak value.

1.2.1 Link to Market Regimes

Market regimes strongly influence drawdown behaviour. Different phases of market activity produce distinct risk characteristics:

- **Growth regimes:** Typically exhibit small and short-lived drawdowns as prices trend upward with relatively low volatility.
- **Crisis regimes:** Characterized by large and prolonged drawdowns caused by sharp price declines and delayed recovery.

1.3 Role of AI in Financial Analysis

The growing complexity of financial markets has driven the adoption of **Artificial Intelligence (AI)** for data analysis and decision-making. Financial time series exhibit nonlinear behaviour and hidden patterns, which are difficult to capture using traditional models. AI methods learn such patterns directly from data without strong prior assumptions. Machine learning algorithms analyse historical market data to identify relationships between returns, volatility, and other variables. Unlike classical approaches, AI models adapt dynamically to evolving market conditions.

A key application is the identification of market regimes. Techniques such as clustering and neural networks detect latent states in financial time series by recognizing similarities in statistical behaviour across time. These methods enable automated classification of market environments by risk and stability.

AI also supports risk management by detecting transitions between stable and turbulent conditions, allowing investors to adjust strategies and reduce exposure to market downturns. AI has thus become essential to modern quantitative finance.

1.4 Objectives of the Study

This study investigates the application of AI techniques to identify market regimes and associated risk patterns, providing insights into how market dynamics evolve over time.

The specific objectives are:

- Analyze financial time series data to examine variations in returns and volatility across different periods.
- Identify distinct market regimes using AI and machine learning techniques.
- Study the relationship between market regimes and investor risk, particularly through drawdown behaviour.
- Evaluate how regime identification supports improved risk management and investment decisions.
- Interpret AI-based results within a financial context rather than purely statistical outcomes.

Through these objectives, the study demonstrates how AI can enhance the understanding of complex financial systems and contribute to adaptive, risk-aware investment strategies.

Chapter 2

Mathematical Background

2.1 Financial Time Series Representation

Financial market data are observed sequentially over time and are naturally modeled as time series. Let P_t denote the price of a financial asset at discrete time points

$$t = 0, 1, 2, \dots, T.$$

The collection

$$\{P_t\}_{t=0}^T$$

describes the evolution of asset prices through time.

Definition 2.1.1. A **financial time series** is a stochastic process $\{P_t\}_{t=0}^T$ in which each observation P_t is treated as a random variable representing the asset price at time t .

Financial prices exhibit uncertainty due to changing economic conditions and market interactions. Consequently, future values cannot be ascertained deterministically and are modeled probabilistically. Moreover, observations are typically dependent

across time, meaning that price movements may be influenced by past information.

Price series often display non-stationary behaviour, including trends and changing variability, which complicates direct statistical analysis. For this reason, financial modelling commonly focuses on asset returns rather than prices. The mathematical definition of returns is introduced in the next section.

2.2 Asset Returns

Financial analysis is commonly performed using returns rather than raw prices, since returns measure relative changes in asset value and are comparable across time.

1. Simple Return

The one-period simple net return is defined as:

$$R_t := R_t(1) = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

2. Multiperiod Return

For an investment held over k periods, where k denotes the number of holding periods, the k -period simple gross return is given by,

$$1 + R_t(k) := \frac{P_t}{P_{t-k}} = \prod_{j=0}^{k-1} (1 + R_{t-j}).$$

3. Logarithmic Return

The logarithmic (log) return is defined as:

$$r_t := r_t(1) = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(1 + R_t).$$

A key property of log returns is additivity across time:

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j} = \ln \left(\frac{P_t}{P_{t-k}} \right).$$

The relationship between simple and log returns is given by,

$$R_t(k) = e^{r_t(k)} - 1, \quad r_t(k) = \ln(1 + R_t(k)).$$

Returns provide a scale-independent representation of asset performance and are therefore widely used in financial time series modelling.

2.3 Volatility and Second-Order Properties

Financial time series exhibit time-varying variability known as **volatility**. Volatility measures the magnitude of fluctuations in asset returns and is central to financial risk modelling. Let $\{r_t\}$ denote the return process.

1. Mean Return

The expected return is defined as:

$$\mu := \mathbb{E}[r_t].$$

2. Variance

The variance of returns is defined as:

$$\sigma^2 = \text{Var}(r_t) := \mathbb{E}[(r_t - \mu)^2].$$

3. Volatility

Volatility is defined as the standard deviation of returns:

$$\sigma := \sqrt{\text{Var}(r_t)}.$$

It measures the typical magnitude of return fluctuations and serves as a primary indicator of financial risk.

4. Sample Volatility

Given observations $r_1, \dots, r_T$, the sample volatility is given by,

$$\hat{\sigma} := \sqrt{\frac{1}{T-1} \sum_{t=1}^T (r_t - \hat{\mu})^2}.$$

where $\hat{\mu}$ denotes the sample mean return defined as :

$$\hat{\mu} := \frac{1}{T} \sum_{t=1}^T r_t.$$

5. Conditional Volatility

Empirical financial returns exhibit volatility clustering, meaning that

$$\text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2$$

changes over time, where $\mathcal{F}_{t-1}$ denotes the information set available up to time $t-1$.

Returns are therefore modeled as follows:

$$r_t = \mu_t + \sigma_t \varepsilon_t,$$

where,

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}(\varepsilon_t) = 1.$$

The time-varying quantity σ_t is called conditional volatility.

2.4 Drawdown and Risk Measures

While variance and volatility measure average fluctuations in returns, investors are often more concerned with downside losses. Drawdown-based measures quantify the decline of a portfolio from its historical peak and provide an intuitive assessment of investment risk.

1. Running Maximum

The running maximum up to time t is defined as:

$$M_t := \max_{0 \leq s \leq t} P_s$$

which represent the highest value achieved by the asset up to time t .

2. Drawdown

The drawdown at time t is defined as follows:

$$D_t := \frac{M_t - P_t}{M_t}$$

This quantity measures the relative decline from the historical peak and satisfies

$$0 \leq D_t \leq 1.$$

3. Maximum Drawdown

The maximum drawdown over a time horizon $[0, T]$ is given by,

$$\text{MDD} := \max_{0 \leq t \leq T} D_t$$

Maximum drawdown represents the worst cumulative loss experienced by an investor before recovery and is widely used as a downside risk metric.

Unlike volatility, which treats positive and negative fluctuations symmetrically, drawdown focuses exclusively on losses. Large drawdowns are typically associated with crisis regimes, whereas stable market regimes exhibit smaller and shorter drawdowns. Drawdown measures therefore provide a direct link between statistical properties of returns and investor risk perception, making them particularly useful in regime-based financial modelling.

2.5 Markov Processes and Regime Representation

Financial markets often exhibit structural changes over time, where the statistical properties such as mean returns and volatility vary across different periods. These changes can be modeled using stochastic processes that evolve between a finite number of states called **regimes**.

1. Markov Property

Let $\{S_t\}$ be a discrete-time stochastic process representing the market regime at time t . The process satisfies the **Markov property** if :

$$\mathbb{P}(S_t = j \mid S_{t-1}, S_{t-2}, \dots, S_0) = \mathbb{P}(S_t = j \mid S_{t-1}),$$

meaning that the future state depends only on the current state and not on the entire past history.

2. State Space

Assume the regime variable takes values in a finite set :

$$S_t \in \{1, 2, \dots, K\},$$

where each state corresponds to a distinct market condition, such as low-volatility or high-volatility regimes.

3. Transition Probabilities

Transitions between regimes are governed by probabilities

$$p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i),$$

which form the **transition probability matrix**

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{pmatrix},$$

where,

$$\sum_{j=1}^K p_{ij} = 1, \quad p_{ij} \geq 0.$$

4. Regime-Dependent Returns

Returns are assumed to depend on the current regime:

$$r_t = \mu_{S_t} + \sigma_{S_t} \varepsilon_t,$$

where ε_t is a random variable with zero mean and unit variance.

Different regimes therefore have different mean returns and volatilities.

The regime variable $\{S_t\}$ is typically unobserved and must be inferred from observed financial data. Modeling markets using Markov processes allows transitions between stable and turbulent periods to be represented probabilistically, forming the mathematical foundation for regime-switching and hidden Markov models used in financial analysis.

Chapter 3

Dataset and Preprocessing

The empirical analysis in this study is based on the **S&P 500 index**, one of the most widely used indicators of overall equity market performance. The index represents large-cap companies listed on stock exchanges in the United States and is commonly employed as a benchmark for studying aggregate market behaviour.

Historical price data were obtained using the `yfinance` Python library, which provides access to financial data available through Yahoo Finance. The dataset spans the period from **January 1, 2001, to January 1, 2025**, covering approximately twenty-four years of daily trading observations.

Notable events within the sample include:

- The dot-com market correction in the early 2000s
- The global financial crisis of 2008
- The COVID-19 market disruption in 2020

	Close	High	Low	Open	Volume
Date					
2001-01-02	1283.270020	1320.280029	1276.050049	1320.280029	1129400000
2001-01-03	1347.560059	1347.760010	1274.619995	1283.270020	1880700000
2001-01-04	1333.339966	1350.239990	1329.140015	1347.560059	2131000000
2001-01-05	1298.349976	1334.770020	1294.949951	1333.339966	1430800000
2001-01-08	1295.859985	1298.349976	1276.290039	1298.349976	1115500000

Figure 3.1: Preview of the S&P 500 dataset showing initial observations

3.1 Feature Construction

Using the cleaned price data, several features were created for market regime detection. These features are based on the mathematical concepts introduced in Chapter 2 and help describe market movement and risk behaviour.

1. Log Returns

Log returns were calculated using consecutive closing prices. Returns measure the relative change in price and are commonly used instead of raw prices in financial analysis.

```
df["return"] = np.log(df["Price"] / df["Price"].shift(1))
df.dropna(inplace=True)
```

2. Volatility

Market volatility was estimated as the rolling standard deviation of returns over a 20-day window. This measure captures how strongly prices fluctuate over time and reflects changing market risk.

```
df["volatility"] = df["return"].rolling(window=20).std()
df.dropna(inplace=True)
```

3. Drawdown

Drawdown was computed to measure the decline of the market from its previous peak. In addition, drawdown duration was calculated to record how long the market remains below its maximum value.

```
df["running_max"] = df["Price"].cummax()
df["drawdown"] = (df["Price"] - df["running_max"]) / df["running_max"]
```

4. Rolling Features

Rolling statistics were constructed over 20-day and 100-day windows to capture short-term and long-term market behaviour. These features help identify persistent trends and variations in volatility.

```
df["rolling_mean_20"] = df["return"].rolling(20).mean()
df["rolling_vol_20"] = df["return"].rolling(20).std()
df["rolling_mean_100"] = df["return"].rolling(100).mean()
df["rolling_vol_100"] = df["return"].rolling(100).std()
```

5. Feature Normalization

All features were standardized using the `StandardScaler` so that each variable has a comparable scale. This step ensures that no single feature dominates the learning algorithm due to differences in magnitude.

```
scaler = StandardScaler()
df_scaled = scaler.fit_transform(df)
df_scaled = pd.DataFrame(df_scaled, columns=df.columns, index=df.index)
```

3.2 Final Dataset for Modeling

The final dataset contains standardized features representing market returns, volatility, drawdown behaviour, and rolling statistics. Each row corresponds to one trading day, and each column represents a feature used for analysis. This processed dataset is used as the input for the Hidden Markov Model applied in the next chapter to identify market regimes.

Date	Price	return	volatility	running_max	drawdown	drawdown_duration	rolling_mean_20	rolling_vol_20	rolling_mean_100	rolling_vol_100
2001-06-22	-0.736406	-0.806692	-0.118009	-0.798208	0.076784	-0.633422	-1.259531	-0.118009	-1.390794	0.658217
2001-06-25	-0.741733	-0.478415	-0.151136	-0.798208	0.035556	-0.631101	-1.124101	-0.151136	-1.389688	0.658158
2001-06-26	-0.743186	-0.146691	-0.164417	-0.798208	0.024317	-0.628780	-0.989712	-0.164417	-1.458629	0.655567
2001-06-27	-0.747677	-0.409266	-0.247921	-0.798208	-0.010437	-0.626460	-0.753609	-0.247921	-1.330393	0.640167
2001-06-28	-0.735735	1.004580	-0.188239	-0.798208	0.081975	-0.624139	-0.620928	-0.188239	-1.242840	0.649668

Figure 3.2: Preview of the processed and normalized dataset used for regime detection

Chapter 4

Market Regime Detection

4.1 Unsupervised Learning for Regime Detection

Market regimes are **unobserved** (latent) states that cannot be directly identified from price data. Unlike supervised learning, where models are trained using the labeled examples, regime detection requires discovering patterns in financial time series without predefined labels. For this reason, unsupervised learning methods are appropriate for the analysis.

Two unsupervised approaches are used in this study:

1. **K-Means Clustering**: Used as an exploratory method to group observations based on similarity of financial features.
2. **Hidden Markov Model (HMM)**: A probabilistic model that captures time dependence and transitions between hidden market states.

K-Means provides an initial segmentation of the data, while the HMM models the sequential structure of financial markets and the persistence of regimes over time.

4.2 K-Means Clustering

K-Means clustering is an unsupervised learning algorithm that divides data into K distinct clusters. Each observation is assigned to the cluster whose centroid is closest, where the centroid represents the mean of all points in that cluster.

1. Algorithm Description

Given observations $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in \mathbb{R}^d$, the K-Means algorithm minimizes the within-cluster variation defined by,

$$\operatorname{argmin}_{\mathbf{C}} \sum_{i=1}^K \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2,$$

where $\boldsymbol{\mu}_i$ denotes the centroid of cluster C_i .

The algorithm proceeds iteratively:

- (a) Initialize K centroids.
- (b) Assign each observation to the nearest centroid.
- (c) Update centroids as the mean of assigned observations.
- (d) Repeat until convergence.

2. Application to Market Data

For regime detection, the standardized feature set consisting of returns, volatility, drawdown, and rolling statistics was used as input. The number of clusters was chosen as $K = 3$, representing three market conditions: growth, transitional, and crisis regimes.

```
from sklearn.cluster import KMeans  
  
kmeans = KMeans(n_clusters=3, max_iter=300, random_state=42)  
  
df["regime"] = kmeans.fit_predict(df_scaled)
```

```
df["regime"].value_counts()
```

3. Cluster Distribution

The number of observations assigned to each cluster is shown in Table 4.1.

Regime	Count
0	2370
1	3027
2	521

Table 4.1: Number of observations per cluster

4. Regime Characteristics

Each cluster was analyzed using the mean values of key financial features.

Regime	Mean Return	Mean Volatility	Mean Drawdown
0	0.000182	0.009378	-0.181732
1	0.000708	0.007969	-0.033694
2	-0.001951	0.025864	-0.306507

Table 4.2: Mean feature values by K-Means cluster

Based on these statistics, the clusters can be interpreted as:

- **Regime 0 (Normal):** Moderate returns and volatility with intermediate drawdowns.
- **Regime 1 (Bull market):** Positive returns with low volatility and small drawdowns.
- **Regime 2 (Crisis regime):** Negative returns with high volatility and large drawdowns.

5. Visualization

Figure 4.1 shows the detected regimes overlaid on the S&P 500 price series.

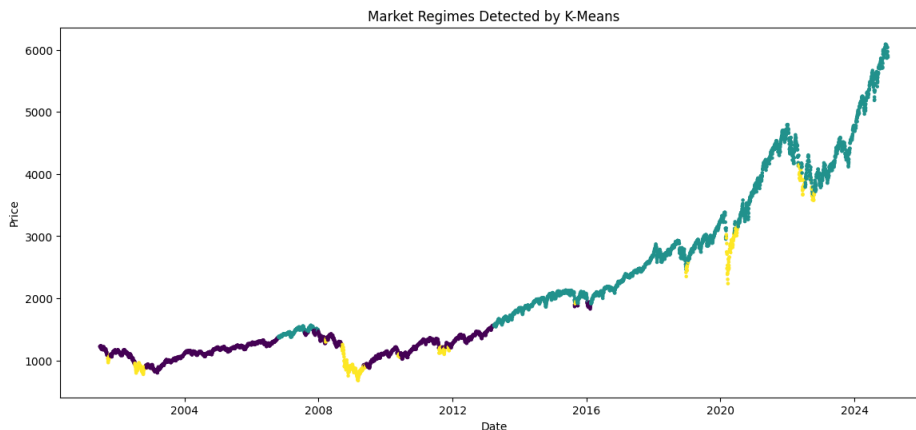


Figure 4.1: Market regimes detected by K-Means clustering

The visualization shows that the crisis regime aligns with the major market downturns, including the 2008 financial crisis and the 2020 COVID-19 market decline, while the bull regime corresponds to sustained growth periods.

4.3 Hidden Markov Model

Unlike K-Means, which provides a static grouping of market conditions, the financial markets evolve over time. Market regimes usually persist for a period before changing. The Hidden Markov Model (HMM), introduced in Section 2.5, is used to capture these time-dependent transitions between hidden market states.

1. Algorithm Description

The HMM estimates the model parameters and hidden states using the following steps:

- (a) **Initialization:** Start with initial estimates for transition probabilities, means, and variances for each state.

- (b) **Expectation Step (E-step):** Compute the probability of being in each hidden state at each time point given the observed returns and current parameter estimates.
- (c) **Maximization Step (M-step):** Update the model parameters (transition probabilities, state means, and variances) to maximize the likelihood of the observed data.
- (d) **Iteration:** Repeat the E-step and M-step until the parameters converge or the maximum number of iterations is reached.
- (e) **State Prediction:** Finally, the model assigns the most probable regime to each trading day using the learned parameters.

2. Model Specification

A Gaussian HMM with three hidden states ($K = 3$) was used to remain consistent with the K-Means analysis. The covariance type was set to **full**, allowing each regime to have its own variance structure.

```
from hmmlearn.hmm import GaussianHMM

hmm = GaussianHMM(
    n_components=3,
    covariance_type="full",
    n_iter=1000,
    random_state=42
)

hmm.fit(hmm_features)
df["hmm_regime"] = hmm.predict(hmm_features)
```

3. Transition Probability Matrix

The estimated transition probability matrix is:

$$P = \begin{pmatrix} 0.9824 & 0.0135 & 0.0042 \\ 0.0204 & 0.9796 & 0.0000 \\ 0.0087 & 0.0000 & 0.9913 \end{pmatrix}$$

The large diagonal values (greater than 0.97) indicate strong persistence, meaning the market tends to remain in the same regime for extended periods.

4. Regime Characteristics

The average feature values for each detected regime are shown in Table 4.3.

Regime	Mean Return	Mean Volatility	Mean Drawdown
0	0.000086	0.009062	-0.102133
1	0.000823	0.006006	-0.007095
2	-0.000104	0.017370	-0.285952

Table 4.3: Mean feature values by HMM regime

The regimes can be interpreted as:

- **Regime 0 (Normal):** Moderate returns and volatility with intermediate drawdowns.
- **Regime 1 (Bull market):** Positive returns with low volatility and small drawdowns.
- **Regime 2 (Crisis regime):** Negative returns with high volatility and large drawdowns.

5. Regime Persistence

The duration statistics of each regime are presented in Table 4.4.

Regime	Mean Duration	Median Duration	Max Duration
0	58.0	20.0	714
1	51.1	36.0	310
2	117.0	47.5	607

Table 4.4: Regime duration statistics in days

The crisis regime shows the longest average duration, indicating that the market recovery periods tend to be slow once a crisis begins.

6. Visualization

Figure 4.2 illustrates the detected regimes overlaid on the S&P 500 price series.

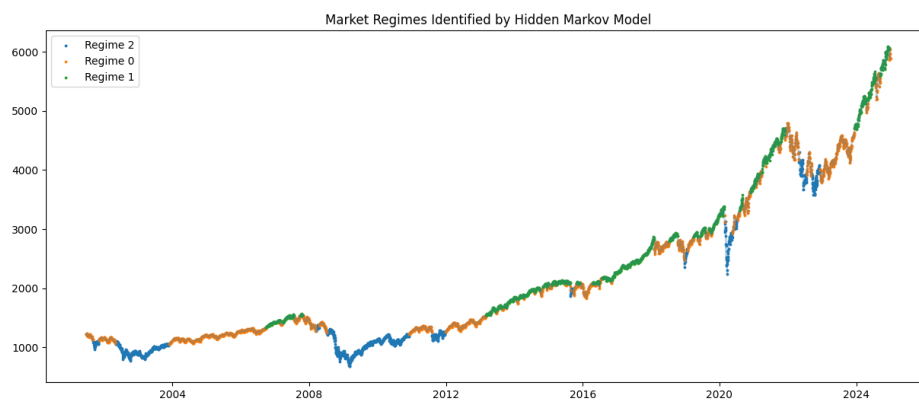


Figure 4.2: Market regimes identified by the Hidden Markov Model

4.4 Summary

Both K-Means and the Hidden Markov Model identified three main market regimes: bullish, normal, and bearish. While K-Means provided an initial grouping of the market conditions, the HMM captured how regimes evolve over time through transition probabilities and duration analysis. The results show that market regimes are persistent and tend to last for extended periods. These identified regimes form the basis for the investment strategy to be developed in Chapter 5.

Chapter 5

Investment Strategy and Evaluation

5.1 Overview

The market regimes identified in the previous chapter provide a structured view of market behaviour. In this chapter, these regimes are used to design a simple investment strategy and evaluate its performance.

A baseline **buy-and-hold** strategy is first considered as a benchmark. This strategy represents a passive investor who remains fully invested in the market at all times.

Next, a **regime-based strategy** is constructed, where investment decisions depend on the detected market regime. Based on the HMM results from Chapter 4, Regime 1 (Bull market) is treated as the favourable state. The strategy invests in the market only when this regime is active; otherwise, it remains in cash.

The performance of both strategies is then compared using total return, volatility, Sharpe ratio, and maximum drawdown.

5.2 Buy-and-Hold Strategy

The buy-and-hold strategy is used as the benchmark. The investor remains fully invested throughout the entire period. The daily return equals the market return, and cumulative performance is obtained by compounding over time.

```
df["buy_hold_return"] = df["return"]
df["buy_hold_cum"] = (1 + df["buy_hold_return"]).cumprod()
```

This strategy represents a passive investment approach and provides a baseline against which the regime-based strategy can be evaluated.

5.3 Regime-Based Strategy

The regime-based strategy adjusts investment decisions based on the detected market regime. Unlike the buy-and-hold approach, this strategy attempts to avoid large losses by exiting the market during high-risk periods.

1. Investment Rule

Based on the HMM results from Chapter 4, Regime 1 (Bull market) is identified as the favourable state characterized by positive returns and low volatility. The strategy follows:

- (a) **Invested:** Invest in the market only during good times (Regime 1).
- (b) **Not invested:** Stay in cash during normal or crisis periods (Regime 0 or Regime 2).

2. Cash Return Assumption

During non-invested periods, a small constant return is assigned to represent a near-zero cash position. The daily cash return is set to 8×10^{-6} , which is approximately a 0.2% annual return.

```

GOOD_REGIME = 1
df["strategy_return"] = 0.000008
df.loc[df["hmm_regime"] == GOOD_REGIME, "strategy_return"] = df["return"]
df["strategy_cum"] = (1 + df["strategy_return"]).cumprod()

```

The strategy follows a simple rule. It remains invested when the market environment is favorable and moves to cash when the market enters neutral or crisis regimes. This reduces losses during downturns and captures gains during bull markets.

5.4 Performance Evaluation

To compare the two strategies, four standard financial metrics are used. Volatility and maximum drawdown were previously defined. The total return and Sharpe ratio are defined below.

1. Total Return

The total return measures the overall growth of an initial investment over the entire period.

$$\text{Total Return} = \frac{\text{Final Wealth}}{\text{Initial Wealth}} - 1$$

2. Sharpe Ratio

The Sharpe Ratio evaluates risk-adjusted performance. It measures return earned per unit of risk:

$$\text{Sharpe Ratio} = \frac{\text{Mean Return}}{\text{Volatility}} \times \sqrt{252}$$

The factor $\sqrt{252}$ annualizes the ratio, accounting for approximately 252 trading days in a year. A higher Sharpe Ratio indicates better risk-adjusted returns.

The metrics are computed as follows:

```
# Total return
bh_total = df["buy_hold_cum"].iloc[-1] - 1
st_total = df["strategy_cum"].iloc[-1] - 1

# Volatility
bh_vol = df["buy_hold_return"].std() * np.sqrt(252)
st_vol = df["strategy_return"].std() * np.sqrt(252)

# Sharpe ratio (annualized)
bh_sharpe = (df["buy_hold_return"].mean() /
             df["buy_hold_return"].std()) * np.sqrt(252)
st_sharpe = (df["strategy_return"].mean() /
             df["strategy_return"].std()) * np.sqrt(252)

# Maximum drawdown
def max_drawdown(cum_returns):
    running_max = cum_returns.cummax()
    drawdown = (cum_returns - running_max) / running_max
    return drawdown.min()

bh_mdd = max_drawdown(df["buy_hold_cum"])
st_mdd = max_drawdown(df["strategy_cum"])
```

5.5 Results

The performance metrics for both strategies are as follows:

Metric	Buy & Hold	Regime Strategy
Total Return	207.8%	337.9%
Volatility (annualized)	19.2%	4.9%
Sharpe Ratio	0.35	1.30
Maximum Drawdown	-61.0%	-4.5%

Table 5.1: Performance comparison

The regime-based strategy performs better than buy-and-hold in all metrics.

1. **Total Return:** The regime strategy delivers 337.9% cumulative return compared to 207.8% for buy-and-hold, an improvement of over 130 percentage points.
2. **Volatility:** The regime strategy exhibits significantly lower volatility (4.9% vs 19.2%), indicating much more stable returns.
3. **Sharpe Ratio:** With a Sharpe Ratio of 1.30 versus 0.35, the regime strategy provides substantially better risk-adjusted returns.
4. **Maximum Drawdown:** The regime strategy limits losses to just -4.5%, while buy-and-hold experiences a significant -61.0% drawdown during the 2008 financial crisis.

5.6 Visual Comparison

Figure 5.1 shows the cumulative growth of \$1 invested in each strategy over the 24-year period.

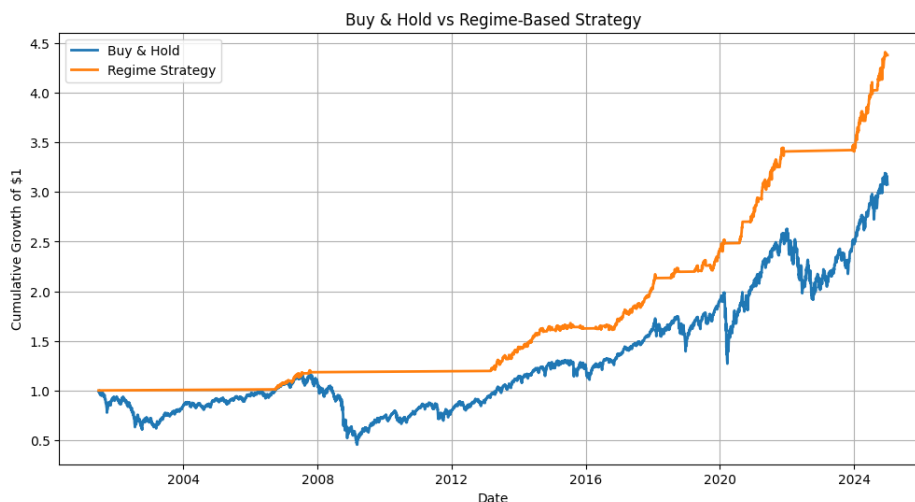


Figure 5.1: Buy-and-Hold vs Regime-Based Strategy

The plot highlights the key difference between the two approaches. The buy-and-hold strategy experiences severe drawdowns during major crises (2008, 2020), taking years to recover. In contrast, the regime-based strategy avoids these losses by moving to cash during crisis regimes, resulting in steady, consistent growth. While the regime strategy grows more slowly during bull markets, its capital preservation during downturns leads to higher overall wealth by the end of the period.

Chapter 6

Conclusion and Future Scope

6.1 Summary of Findings

This study successfully detected three distinct market regimes in the S&P 500 index from 2001 to 2025 using K-Means clustering and a Hidden Markov Model. The Hidden Markov Model revealed strong regime persistence, with diagonal transition probabilities above 0.97, confirming that market conditions tend to last over time. Crisis regimes were found to have the longest average duration (117 days), indicating slow recovery periods.

A simple regime based strategy, investing only during the bull regime and staying in cash otherwise, was evaluated against a buy and hold benchmark. The regime strategy delivered higher returns, lower volatility, a better Sharpe ratio, and significantly smaller drawdowns, demonstrating the value of regime aware investing.

6.2 Limitations

This study has a few limitations:

1. **Only one index:** The analysis uses only the S&P 500. Results may not hold

for other markets.

2. **Simple strategy:** The strategy only chooses between investing in the market or staying in cash. It does not consider partial investment or other assets.
3. **No trading costs:** The analysis ignores transaction costs, taxes, and liquidity issues. In reality, these would reduce profits.
4. **No out-of-sample testing:** The model was tested on the same data used for training. Future performance could be different.
5. **Fixed number of regimes:** The number of regimes (3) was chosen in advance. The optimal number may vary.

6.3 Future Scope

Several directions can be explored in future work:

1. **Multiple assets:** Apply the same approach to other indices, sectors, or individual stocks to see if regimes are similar across markets.
2. **Better strategies:** Instead of just investing or staying in cash, the strategy could adjust how much to invest based on regime probabilities.
3. **Including economic data:** Add macroeconomic variables like interest rates or inflation to improve regime prediction.
4. **Testing on new data:** Validate the model on data after 2025 to check if the strategy still works.
5. **Other models:** Try different machine learning methods such as neural networks or Markov-Switching GARCH for comparison.

6.4 Concluding Remarks

This study demonstrates that market regimes can be effectively detected using the unsupervised learning methods. The Hidden Markov Model successfully identified three distinct market states: bull, normal, and crisis.

A simple strategy based on these regimes, which invests only during bull markets and stays in cash otherwise, outperformed the traditional buy-and-hold approach. It delivered higher returns, lower risk, and much smaller losses during market downturns.

Although the study has limitations, the results suggest that regime aware investing is a promising approach. With further refinement and testing on new data, such methods could become valuable tools for investors and risk managers.

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